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About

Long-Kai Huang is currently an Assistant Professor in the Department of Computer Science at Hong Kong Baptist University. Prior to joining HKBU, he was a Senior Researcher at Tencent AI Lab. He received his Ph.D. in Computer Science and Engineering from Nanyang Technological University (NTU), Singapore, under the supervision of Prof. Sinno Jialin Pan.

His research focuses on the foundational theory and applications of machine learning. He is also interested in AI for science. He has published more than 30 papers in top machine learning and AI conferences and journals, including ICML, NeurIPS, ICLR, TKDE, TNNLS, and Nature Machine Intelligence. He received the Outstanding Paper Award Honorable Mention at ICLR 2024 for his work on meta-continual learning, as well as the ICLR Outstanding Reviewer Award.

Research Interests

- Continual Learning, Meta-Learning and Transfer Learning
- Efficient Pre-training and Post-training for LLMs
- AI for Science: AI-Aided Drug Discovery and Single Cell Omics
- Optimization, Learning Dynamic

Selected Publications

- **LK Huang**, R Zhu, B He, J Yao. “Steering Protein Language Models.” In International Conference on Machine Learning (**ICML**), 2025
- Z Mo, **LK Huang**, SJ Pan. “Parameter and Memory Efficient Pretraining via Low-rank Riemannian Optimization.” In International Conference on Learning Representations (**ICLR**), 2025
- Y Wu*, H Wang*, P Zhao, Y Zheng, Y Wei, **LK Huang**†. “Mitigating Catastrophic Forgetting in Online Continual Learning by Modeling Previous Task Interrelations via Pareto Optimization.” In International Conference on Machine Learning (**ICML**), 2024
- Y Wu, **LK Huang**†, R Wang, D Meng, Y Wei. “Meta Continual Learning Revisited: Implicitly Enhancing Online Hessian Approximation via Variance Reduction.” In International Conference on Learning Representations (**ICLR**), 2024.
- S Zhou, X Huang, N Liu, H Zhou, FL Chung, **LK Huang**†. “Improving Generalizability of Graph Anomaly Detection Models via Data Augmentation.” IEEE Transactions on Knowledge and Data Engineering (**TKDE**). 35(12), pp.12721-12735. 2023.
- **LK Huang**, J Huang, Y Rong, Q Yang, Y Wei. “Frustratingly Easy Transferability Estimation.” In International Conference on Machine Learning (**ICML**), 2022

- H Yao*, **LK Huang***, L Zhang, Y Wei, L Tian, J Zou, J Huang. “Improving Generalization in Meta-learning via Task Augmentation.” In International Conference on Machine Learning (**ICML**), 2021
- **LK Huang** and SJ Pan. “Communication-Efficient Distributed PCA by Riemannian Optimization.” In International Conference on Machine Learning (**ICML**), 2020